

## Business findings & Recommendations

1. Approximately 2/3 of the orders are “multi-item”, meaning there are more than 1 type of pizza within that order. As expected, the majority of the revenue comes from “multi-item” orders. Therefore, it is in the company’s best interest to incentivize the customers into buying multiple types of pizza within a single order. Doing so will help increase the average order value and overall revenue. The recommendations for achieving this are as follows:

- Offering different combos such as a Large Barbecue Chicken Pizza and a Small Thai Chicken Pizza for only \$30 instead of \$33.50. However, identifying the most effective combinations would require further analysis of pizza co-purchase patterns.
- Speaking of combos, an alternative but similar way is to offer discounts to specific pizzas to stimulate the ordering of different types of pizza. Taking the example above, a customer might be ordering a Large Barbecue Chicken Pizza but notices that the Small Thai Chicken Pizza is currently having a 20% discount. As a result, they might feel more compelled to getting an extra pizza for the sake of variety and a good bargain. The choice of which pizza to be discounted depends on the purpose of the company. If the company’s intention is to improve sales within a downturn period, then it is better to offer discounts to highest-selling pizzas such as The Classic Deluxe Pizza\* (See the top 5 pizzas table and the quadrant chart in the Pizza page for more information). Conversely, if the company is trying to clear out a specific category of pizzas for reasons such as the ingredients use to make them is in stock for too long and risk being spoiled, then discounts could be offered to those specific categories of pizzas. The specifics for how much to discount require further price analysis to ensure profit is still being maintained.
- While looking through the dataset, I can only see details about pizzas. Therefore, it is my assumption that the company currently only sells pizzas. The important thing to take note of is that a pizza manufacturing and distribution company does not necessarily need to only sell pizzas. In fact, by restricting the menu to only pizzas and excluding complementary side dishes such as salads, appetizers (chicken/buffalo wings, fries, etc), and sodas, we are limiting the customers’ additional purchase options to go alongside their pizza. The intention is not to include all appetizers and side dishes mentioned above, but to at least cover the most common ones such as sodas. This will boost the average order value, and in turn the total revenue as well. The hypothesis is that a customer that is buying a pizza would be more inclined to buy a soda or fries along with it.

2. Looking at the relationship between pizza size and total quantity sold across the different pizza categories, it can be seen that customers frequently buy Large (L)-size pizzas. This is a good thing since it generates better revenue. However, the same trend does not apply to pizzas within the Classic category as the majority of the customers seemingly prefer to order the Small (S)-size pizzas instead. The recommendation is to try and encourage the customers to upgrade to larger-sized pizzas within the Classic category. One way to achieve this is to alter the pricing strategy between the different sizes. Take for example the Hawaiian pizza and the price of the different sizes below:

| Pizza                   | Price   |
|-------------------------|---------|
| Hawaiian Pizza (Small)  | \$10.5  |
| Hawaiian Pizza (Medium) | \$13.25 |
| Hawaiian Pizza (Large)  | \$16.5  |

The current pricing follows a roughly linear progression between sizes. This may provide relatively little psychological incentive for customers to upgrade from Small to Medium/Large. They will likely continue to stick with Small-size pizzas due to consumer psychology of it being the cheapest. Nevertheless, we can also leverage the consumer psychology and skew the pricing across the different sizes to make it more non-linear. The impact of this action is to shift the customer's mindset from "cheapest is more cost-effective" to "it doesn't take much to upgrade to a bigger size". The Hawaiian Pizza example has a price increase of approximately \$3 for each size, but if we change it to the following price below, then the customers might be more willing to change to a Medium-size pizza. Once again, the specifics on what price to change to requires further price analysis. Nevertheless, it is recommended to initially keep the Small price unchanged. This is to avoid negatively affecting the perceived affordability of the most common option.

| Pizza                   | Price  |
|-------------------------|--------|
| Hawaiian Pizza (Small)  | \$10.5 |
| Hawaiian Pizza (Medium) | \$12.5 |
| Hawaiian Pizza (Large)  | \$16.5 |

3. The average time between consecutive orders can seem quite high, sitting at an average of 24.58 minutes. This means, on average it takes around half an hour to get another order. However, this is not too concerning when taken into consideration the time difference between the last order of day X and the first order of day X + 1 is extremely long due to closing time. The presence of these outliers can distort the actual value. For example, as seen in the 2 orders in the table below, taken directly from the dataset, the time difference between the closing order on the 1<sup>st</sup> Jan 2015 and the opening order on the 2<sup>nd</sup> Jan 2015 is

806 minutes. Therefore, this average metric should be recalculated using only consecutive orders within the same operating day.

| Order ID | Date (mm-dd-yyyy) | Time     |
|----------|-------------------|----------|
| 69       | 01/01/2015        | 10:12 PM |
| 70       | 01/02/2015        | 11:38 AM |

4. Total orders and revenue generally increase throughout the weekdays, peaking on Friday, before declining over the weekend, with Sunday being the lowest. The identified period of downturn is during the last 4 months of the year (Sep – Dec). However, with only 1-year worth of data, it is not possible to conclusively determine whether the period between September and December is a recurring seasonal trend or not. Nonetheless, all the points mentioned in bullet point 1 above can be considered and apply during these periods of time.